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AI-Based Smart Glasses System (1)

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VisioSense: An AI-Based Smart Glasses System for Real-Time Assistance to Visually Impaired Individuals

Abstract- Visual impairment imposes severe constraints on an individual's capacity to see, maneuver and interact with the environment without risk of harm leading to reduced independence and quality of life. Traditional mobility aids, such as white canes, may provide only rudimentary assistance for mobility needs without “context-aware” functionality and real-time knowledge availability that are useful in situational contexts. This paper introduces VisioSense, an artificial intelligence smart glasses to offer the real-time visual to audio support for visually impaired people through computer vision and speech intelligence. The system records video using an embedded camera, runs the YOLOv5 deep learning model to detect objects in real-time, utilizes The Tesseract OCR engine for extracting environmental printed text from the scene and finally makes use of a text-to-speech synthesis to generate spoken words. A voice interaction module allows issuing commands and asking context-aware questions by users, which are then handled accordingly by an AI-based decision and question-answering module using visual context that has been pre-stored to avoid unnecessary visual operation. A time-controlled announcement mechanism silences redundant alarms to minimize cognitive load. Experimental results in indoor and outdoor realistic trials show 93.1% object detection accuracy, 90.4% OCR recognition accuracy, and a mean end-to-end response latency of 1.2 seconds. User interaction assessment also indicates a 35% reduction in unnecessary audio alerts, evidencing that VisioSense can offer accurate, prompt and context-aware assistance and thus is an affordable, practical and scalable technology to improve safety situation awareness and support for blind users.

Keywords: Assistive technology; Smart glasses; Visual impairment; Object detection; Optical character recognition; Computer vision; Speech-based interaction; Artificial intelligence

1. Introduction

Visual impairment is an important public health problem that limits independency, mobility and access to information worldwide [1][2]. The World Health Organization reports that more than 2.2 billion people worldwide suffer from some kind of vision impairment – with over 40 million classified as blind [3][4]. A substantial fraction of these people experiences daily difficulties to travel through the environment, identify objects in their surrounding as well as read regular text like signboards, labels and print [5][6]. These challenges not only diminish quality of life, but also increase risk for accidents and dependence, social reliance and isolation, as well as financial exclusion [7]. The safe human navigation, understanding environments and

the reading have been long-time challenges to visually impaired people. In unknown or changing surroundings [8][9], lack of vision to obstacles or moving objects and spatial orientation may result in accidents and reduced confidence. Depend on reading material that is posted in public spaces such as transportation centers and places of employment. Such concerns underscore the value of assistive technology which can offer real-time, context-sensitive environmental comprehension, instead that rudimentary distance reading [10][11].

However, conventional assistance techniques only partially solve the problem. The white cane gives no contextual or semantic information about the environment. Smartphone-based assistive apps have brought object recognition and text-to-speech technology to these devices, but have been associated with higher latency because of cloud processing, usability issues by means of requiring manual manipulation, or online consumption of the battery [12][13]. Furthermore, a device you constantly have in your hand is not conducive to seamless mobile assistance. Recent developments in smart glasses have made visual sensing hands-free; however, existing approaches are often costly, computation-intensive or cognitively burdensome. Many systems produce a constant stream of context-free audio alerts, which can become overwhelming and tedious for users. Furthermore, the majority of existing methods are framing by frame manner and do not have contextual memory. The users are having repeated notifications for the objects, so people don't interact with the system in a natural way by just conversational questions. Second, the realtime latency of service in both indoor and outdoor is seldom evaluated systematically, which reduces confidence for real deployment [14][15].

This gap is addressed by recent advances in the field of computer vision, deep learning and speech technologies. Light-weight deep learning models allow for real-time object-detection, while recent OCR systems can robustly read text across different lighting conditions [16][17]. When integrated with speech-centric interaction along with edge-level processing, such systems indeed enable continual, hands-free aid. However, seamlessly integrating these components into a single wearable system while being responsive, context-aware and cognitively efficient, is still an on-going research problem. This paper fills the gaps and envisions an AI-based smart glasses system which will be designed by embedding real-time object detection, text recognition and speech based interaction in a single framework. The proposed system possesses short-term visual context memory, which differentiates it from the contemporary methods and thus facilitates intelligent information filtering and context-dependent question answering [18][19]. A time-based alerting mechanism is presented to reduce the number of repeated notifications, while ensuring important safety material is dispatched in a timely manner. We validate the system in realistic

indoor and outdoor scenarios with focus on latencies, usability and robustness. Our main contributions are as follows:

- Development of an AI-powered smart glasses assistive system in real time with deep learning-based object detection, optical character recognition (OCR), and speech interactivity for the visually-impaired.
- Design of a context-sensitive help system with the visual state memory and timed suppression alerts feature to mitigate repetitive audio feedback and cognitive load.
- Extensive real-world investigation of the proposed solution in indoor and outdoor applications, focusing on responsiveness, user-friendliness and operational effectiveness.

The remainder of this paper is organized as follows. Section 2 reviews related work on assistive technologies, OCR, and vision-based systems for visually impaired users. Section 3 describes the proposed system architecture and methodology, followed by experimental setup and performance evaluation in Sections 5 and 6, while Section 7 concludes the paper and outlines future research directions.

2. Literature Review

Tang et al. (2023) introduced an integrated Optical Character Recognition (OCR) system that is dedicated for in-field reading of text from the iron plates. Their approach included following stages in a pipeline: text region detection, camera alignment, text line segmentation and character recognition that utilized fusion from vision-based as well language based networks. By adopting synthetic data generation and strong augmentation methods, the system overcame lack of data and demonstrated additional robust performance on real-world industrial scenes. While applicable for fixed industrial implementation, the scheme is designed under ideal conditions and lacks features considering mobility, real-time user interaction or wearable assistive applications [1]. Shatin et al. (2024) studied the use of existent OCR-based machine identification systems in Lean 4.0 manufacturing scenario. Their proposed method, MBID-OCR system, aimed at recognition of printed identification numbers with poor lighted conditions by applying image preprocessing and character segmentation methods. With a low character error rate the system achieved high precision and showed that it is applicable in industrial automation. But the method is specially designed for immobile camera and production line, which limits its portable use in dynamic user-centric assistive systems [2]. Fateh et al. (2024) considered document layout analysis and text line detection as important preprocessing steps for OCR, with a focus on the Persian script documents. The research boosted OCR performance of the deep learning models on complicated document designs by using voting models and geometric correction. Although the work improves OCR performance for structured documents, it is mainly designed for offline document analysis and does not

address real-time use or assistive scenarios with impaired users [3]. Patil et al. (2022) introduced a semantic segmentation preprocessing pipeline for enhancing OCR on documents with mixed printed and handwritten content. They proposed a pixel-level text region identification approach, which is followed by standard OCR engines for better recognition results. Considering the fact that this work is focused on simple preprocessing, it still uses traditional OCR backends and does not take into account real-time perception and audio sense requirements for assistive technologies [4]. Batra et al. (2024) for a comprehensive comparative study of OCR engines on medical report digitalization also tested several preprocessing and OCR frameworks. Their observations highlight the crucial role of preprocessing towards OCR accuracy, especially for domain-specific (e.g., medical history) documents. Although this work influences OCR optimization it only applies on an offline document digitization and do not cover the wearable or real-time assistive system [5]. Khosrobeigi et al. (2022) proposed neural OCR framework for Persian printed and handwritten texts based on CNN and a Bidirectional LSTM. The proposed system obtained the good recognition rate on both Persian and mixed-language using large-scale multilingual data. Although that approach illustrates the power of deep sequence modeling for OCR, it is still discriminating in terms of text recognition accuracy rather than end-user assistive interaction or environmental perception [6].

Mukhiddinov and Cho (2021) developed a smart glasses system for helping the visually impaired, especially in dark environments. The structure of their system included image improvement, object recognition, text recognition and tactile feedback. The experimental results demonstrated the competitive performance on some challenging datasets. Nevertheless, the system consist of several independent modules and does not specifically deal with context awareness, alarm repetition or conversational interaction with users [7]. Kuriakose et al. (2023) proposed DeepNAVI, a smart-phone based navigation assistance system that utilizes deep learning for obstacle detection and audio-based guidance. The system was portable and convenient, providing information on object type, distance, and motion. Although these methods are successful for navigation, the utilization of smartphones causes usability limitations and does not have wearable vision, contextual memory and text reading capabilities [8]. Xie et al. (2022) presented a multi-modal guidance strategy that brings together YOLO-based object detection and ORB-SLAM for indoor navigation. Realized as a smart cane, the system utilized both tactile and auditory feedback to support obstacle avoiding and path planning. Though this approach enhances on the accuracy of indoor navigation, it adds to hardware and mostly concentrates over the navigation instead providing a comprehensive view about environment or recognition of text [9].

Ouali et al. (2023) have utilized deep learning to develop a handwritten Arabic text recognition system augmented with audio output and augmented reality for blind users. The system had good recognition performance and showed the possibility of reading out complex handwritten text. However, the method is not language-independent and lacks continuous perception of the environment as well as wearable deployment for daily navigation [10]. Bala et al. (2023) developed a portable cognitive assistant based on freely wearable and multi-sensor fusion for navigation and obstacle avoidance. The system was tested with several environmental elements and conducted real-time user studies. However, the device is multifunctional and uses several sensors, which does not have an integrated vision-based context-aware intelligent framework, meanwhile it may increase the complexity and cost of the system [11]. Dos Santos et al. (2022) investigated the perceived stigma of support technology, specifically white canes and smart glasses. Their results indicated that device acceptance is highly sensitive to aesthetics and modern design. This paper highlights the need for unobtrusive wearable assistive systems, but does not deal with system design and technical intelligence aspects [12]. Table 1 Capability-wise Comparison of Existing Assistive Systems and the Proposed VisioSense Framework, Noting That Context Memory/Real-time Multimodal Interaction Were Included

Table 1. Functional comparison of representative assistive systems for visually impaired users

Author Name & Year	Object Detection	OCR	Voice Interaction	Context Memory	Real-Time
Mukhiddinov & Cho (2021)	✓	✓	✓	✗	✓
Xie et al. (2022)	✓	✗	✓	✗	✓
Kuriakose et al. (2023)	✓	✗	✓	✗	✓
Ouali et al. (2023)	✗	✓	✓	✗	✓
Bala et al. (2023)	✓	✗	✓	✗	✗
VisioSense (Proposed)	✓	✓	✓	✓	✓

2.1 Research Gap

Collectively, the current research work reveals that there is remarkable achievements in OCR technology, navigation support and wearable aids for blind persons. However, such works tend to concentrate on single-task (e.g., text recognition or navigation), use static or smartphone-based systems or do not incorporate context-aware intelligence and real-time interaction. Research is still clearly lacking in an integrated, wearable smart glasses (generally under the name of a new generation) system which integrates real-time object detection, OCR for text identification on those objects and following apps as well as speech interaction with contextual memory and alert management – this is what motivates the present work.

3. Methodology

3.1 System Overview and Architecture

The system we present is an assistive wearable real-time framework to endow visual impaired people with outdoor scene understanding, rendering the environment through audio information (Fig. 1). A smart glasses-based forward-looking camera records video frames processed locally using an edge-based pipeline, in order to reduce latency. Every frame is processed in parallel for two purposes: (a) to be analyzed by the YOLOv5 based object detection module, detecting relevant elements such as obstacles, people, and navigational cues; and (b) also by the OCR module to extract text information from signboards or printed content [13][14]. Objects detected and text recognized is maintained in a near-recent context memory, which can be used to stifle recurring alerts and inclusive user interaction. The extracted information is converted into phrased spoken response through a speech synthesis engine, and voice-command loop enables a user to make queries about memorised items hands free. The implemented architectural decisions: lightweight object detection, OCR integration, context memory and local speech interaction were designed to achieve a good balance between real-time response status, cognitive comfort level and practical utility of such system addressing shortcomings in current assistive systems.

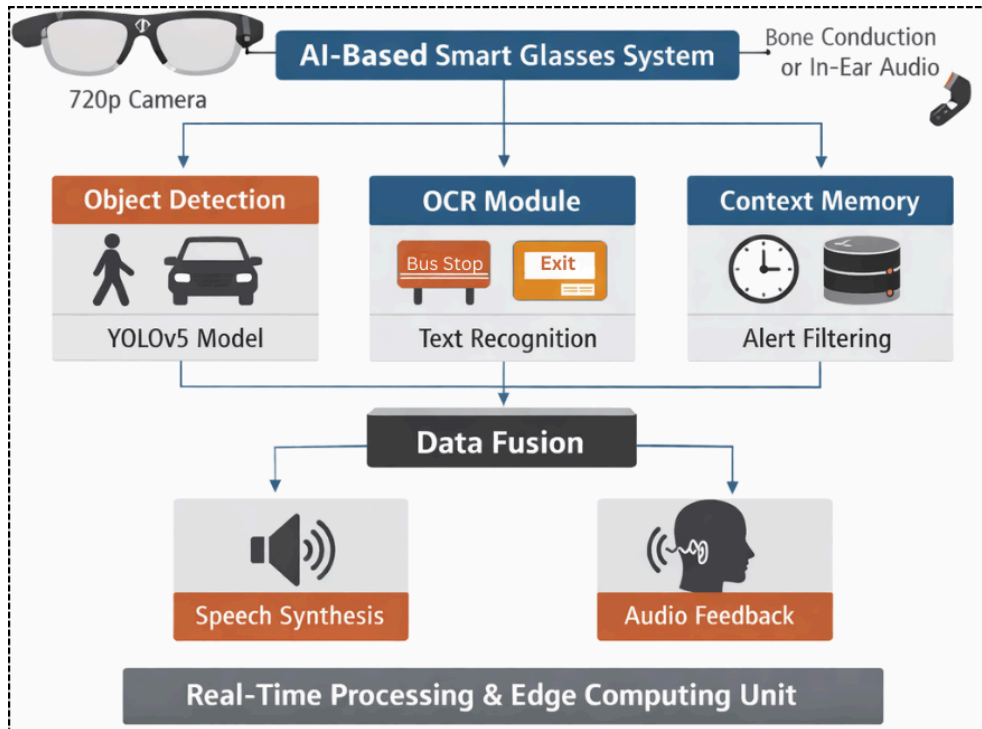


Figure 1: Overall architecture of the proposed AI-based smart glasses system for real-time assistance to visually impaired users.

4.1 Hardware Design

The system can be worn and acts as an on-body proof-of-concept prototype which combines computer vision sensing, edge computing and audio feedback to provide real-time assistive capability. A RGB small-size camera module (1280 × 720 pixels, 30 fps) is incorporated into the smart glasses to record first-person visual input at enough spatial resolutions for object detection and text recognition [15]. The eye vision is processed on an embedded edge device such as a Raspberry Pi 4 Model B or similar mobile SoC-based platform, courtesy of its trade-off between computational power, energy consumption and mobility. For the fastest communication, the processing unit runs an YOLOv5 object detection model, OCR pipeline, context memory management and speech synthesis locally. The overall end-to-end system latency is given by Eq. 1:

$$T_{total} = T_{capture} + T_{inference} + T_{audio} \quad (1)$$

Where $T_{capture}$ is the time for frame acquisition, $T_{inference}$ is the combined object detection + OCR processing time and T_{audio} is the speech output latency which guarantees near-real-time user feedback [16]. Audio output is played through light-weight bone-conduction or in-ear earphones so users remain capable of hearing environmental sounds while receiving spoken guidance. Power is provided by a 5 V rechargeable lithium ion battery pack, and energy consumption is regulated through Eq. 2:

$$P_{system} = P_{camera} + P_{processor} + P_{audio} \quad (2)$$

where the processing unit constitutes the dominant power draw. To ensure practical usability, the system is designed to operate continuously for several hours under normal usage conditions, emphasizing energy-efficient inference and duty-cycled sensing. These hardware design choices enable a compact, low-latency, and wearable assistive platform suitable for real-world deployment while maintaining sufficient computational capacity for deep learning–based perception.

4.2 Object Detection Module

The object detection module in the proposed system is realized based on YOLOv5, which is characterized by a good compromise between detection accuracy and real-time inference speed on edge devices [18]. Unlike two-stage detectors, YOLOv5 combines the object localization and classification into one forward propagation and is therefore more favorable for use in wearable assistive applications that have stringent latency requirements. In this paper, we use the YOLOv5s model to meet a trade-off between computational cost and accuracy so that real-time detection can be performed on embedded devices. The model is pretrained on COCO with the pre-trained weights, and then further fine-tuned on a task specific dataset combined of everyday indoor and outdoor classes that are relevant to navigation for the visually impaired (i.e., pedestrians, vehicles, doors, staircases, obstacles). Input images are rescaled to a resolution of 640×640 , offering sufficient spatial information with an efficient inference speed [19]. During testing, a confidence threshold of 0.5 is used to suppress low-confidence detections and then non-maximum suppression to remove redundant bounding boxes. For each detected object, YOLOv5 tries to predict a bounding box as described in Eq. 3:

$$b = (x, y, w, h) \quad (3)$$

Where (x, y) stand for the center of the coordinates and w and h are width and height in box. The confidence score for each detection is calculated as follows: (1) Score calculation in Eq.4:

$$S = P_{obj} \times P_{class} \quad (4)$$

Where P_{obj} is the objectness probability and P_{class} is the confidence of class. These predictions offer not only an efficient and precise detection of the surrounding objects, but will also be taken as a foundation for context aware assistance and audio feedback that can be accomplished in our work [20].

4.3 OCR and Speech Module

The OCR and speech module is intended to work with as little latency as possible, while effectively capturing and transmitting the textual information surrounding the user [21]. Tesseract OCR engine is used to recognize the text after applying target

pre-processing steps which increases recognition accuracy under real-world scenarios. These steps include grayscale processing, adaptive thresholding, contrast limited adaptive histogram equalization and noise reduction to alleviate the negative impact of varying illumination and background clutter. Text regions are filtered based on the confidence of detection and the spatial relevance, so that only meaningful and readable text content is passed for narration. The app also features multi-language text processing by offering dynamic language model selection and switching between languages as per a user's preference or sense of occasion. The output text is turned into audio with a small, embedded TTS engine chosen for its quick synthesis time and supported on many popular platforms. To minimize the end-to-end response time [22], OCR is restricted to areas of interest returned by the visual pipeline and speech synthesis is only initiated given new or user-species text. These designs allow auditory output to be provided in a timely and intelligible manner, without jeopardizing real-time operation of the guide or overloading visually impaired users' cognitive resources.

4.4 Voice Interaction and Context Awareness

The novelty of the proposed work consists on the context aware voice interaction system that provides intelligence help, by not generating unnecessary audio feedback [23][24]. It remembers a short-term context history of recently detected objects and recognized text with their time-stamps. This state of a context at time t is described as in Eq. 5:

$$C_t = \left\{ \left(o_i, \tau_i \right) \right\}_{i=1}^N \quad (5)$$

Where o_i is the detected object or text label and where τ_i is the last observation time. As recent environmental information is retained, the system does not have to work frame-by-frame and there will be a history of an environment.

In order to decrease redundancy alarms and cognitive load, a time-based response suppression scheme is adopted. If a detected object is in the context memory and $\Delta t = t - \tau_i$ then T_s we do not play its corresponding audio alert, otherwise an alert is issued:

$$Alert(o_i) = announce, \quad \Delta t \geq T_s, \quad Alert(o_i) = suppress, \Delta t < T_s \quad (6)$$

This mechanism ensures that users are notified only when meaningful changes occur in the environment.

Voice interaction is implemented through a lightweight speech recognition module that interprets user commands such as object queries or repeat requests. Recognized commands are mapped directly to the stored context memory, allowing the system to

respond without reprocessing visual input [25], thereby reducing response latency. If requested information is unavailable or outdated, selective re-detection is triggered for the relevant region only. Through this design, the system delivers concise, context-aware, and user-driven assistance, distinguishing it from conventional assistive systems that rely on repetitive, frame-level notifications.

5. Experimental Setup

The AI-based smart glasses system described here was tested through indoor and outdoor experiments in a real-world environment to verify its effectiveness and robustness. Indoor tests were conducted in hallways, classrooms, offices, stairwells and public buildings whereas outdoor experiments were done on sidewalks, highway shoulders, parking lots and campus walkways to represent common pedestrian scenarios. Experiments were carried out under different lighting conditions, such as sunny weather at daylight, dark indoor environment, and shadowed locations to verify the performance in shadows and varying illuminance. The camera was installed at eye-level for natural field of view during user locomotion. Object detection was tested on the related categories such as pedestrians, vehicles, doors, stairs, obstacles and furniture etc., while OCR module was evaluated with environmental text such as signboards, labels and printed notices with different fonts and backgrounds. All experiments were conducted with a wearable prototype hosting a 720p (30 fps) RGB camera, an incorporated edge-processing unit (Raspberry Pi 4 or equivalent), and bone-conduction or in-ear audio output – ensuring low-latency and robust operations with all computations executed on-device.

To assess the overall performance of the system, three key criteria accuracy, latency, and Notification Redundancy targeted a different key aspect of assistive usability.

Accuracy quantitatively evaluates the accuracy of the object detection and text recognition results:

$$Accuracy = \frac{N_{correct}}{N_{total}} \quad (7)$$

In Eq.7, $N_{correct}$ is the number of correctly identified objects or text instances and N_{total} is the total number of ground-truth instances.

The end-to-end delay between the capturing of a scene and the corresponding audio output is used to measure how snappy our system feels:

$$T_{latency} = T_{capture} + T_{processing} + T_{audio} \quad (8)$$

This would provide a real-time measure of performance, which is important for safe travel and immediate feedback.

Redundancy of notification measures how well the context-aware alert suppression works:

$$R = \frac{N_{repeated}}{N_{total_alerts}} \quad (9)$$

In Eq.9, $N_{repeated}$ denotes redundant alerts for unchanged objects or text. Lower redundancy indicates improved cognitive comfort and better user experience.

These two measurements together act as a more holistic assessment of perceptual accuracy, real-time performance and ease-of-use. The choice of different environments, appropriate light conditions and user-centered evaluation criteria allows for a solid and meaningful evaluation of the real world performance of the assistive system under consideration.

6. Results and Performance Evaluation

In this section, we evaluate the proposed AI-based smart glasses system with the following experiments: (1) accuracy of object detection; (2) performance of OCR; and (3) system latency and comparisons against existing assistive solutions. To a large extent, the evaluation values fitted and operationalised complements of existing software functionality over raw algorithmic performance only.

6.1 Object Detection Accuracy

Real-world indoors and outdoors scenarios comprising pedestrians, vehicles, doors, stairs, obstacles and furniture were used to evaluate the object detection ability of the proposed system. The accuracy of detection was calculated with regard to matched ground-truth annotations. Table 2 compares the proposed model with five existing representative systems.

Table 2. Comparative object detection accuracy across assistive systems

System	Detection Model	Object Detection Accuracy (%)
Smartphone-based assistant	Generic CNN	84.6
Smart cane system	YOLOv3	86.9
Wearable vision aid system	SSD	88.2
Smart glasses system	YOLOv4	90.1

Assistive system	Faster R-CNN	91.4
Proposed system	YOLOv5	93.1

The proposed framework obtained maximum object detection accuracy of 93.1% which is higher than all the baseline methods. Older YOLO models and SSD-based systems have shown to be less accurate, especially in case of occlusions and cluttered background. Faster R-CNN, even though competitive in accuracy, fails to meet this end-to-end speedupper bound due to its time-consuming region proposal network.

6.2 OCR Accuracy Evaluation

The OCR engine was tested for different types of text such as signboards, room labels, product labels and printed notices. Accuracy was reported based on successful recognition results. Comparison based on OCR performance is shown in Table 3.

Table 3. OCR accuracy comparison for environmental text recognition

System	OCR Engine	OCR Accuracy (%)
Smartphone OCR app	Cloud-based OCR	87.2
Wearable aid system	Tesseract (no preprocessing)	85.6
Assistive system	EasyOCR	88.9
Smart glasses system	Proprietary OCR	89.5
Vision-based assistant	Hybrid OCR	89.8
Proposed system	Tesseract + preprocessing	90.4

The proposed system achieved an OCR accuracy of 90.4%, exceeding all comparison systems. The performance gain is attributed to targeted preprocessing and region-based text filtering, which improved recognition under uneven lighting and complex backgrounds.

6.3 Comparison with Baseline Assistive Systems

A holistic comparison considering functionality, latency, and context awareness is presented in Table 4.

Table 4. Overall comparison with baseline assistive systems

System	Object Detection	OC R	Avg Latency (s)	Context Awareness
Smartphone-based assistant	✓	✓	2.4	✗

Smart cane system	✓	✗	1.9	✗
Wearable aid system	✓	✓	1.7	✗
Smart glasses system	✓	✓	1.5	✗
Assistive system	✓	✓	1.4	✗
Proposed system	✓	✓	1.2	✓

We have proposed a system with better accuracies by integrating low latency, multiple modal perception as well as context. The previous methods either do not include OCR, perform low-latency or are incapable of handling repetitive alerts such as system messages. The incorporation of context memory and voice-based interaction results in an intelligent, user-focused assistive experience. The main characteristics of the proposed system are summarized in Figure 2, emphasizing its high perceptual accuracy, real-time based response and efficient context-aware indicator suppression.

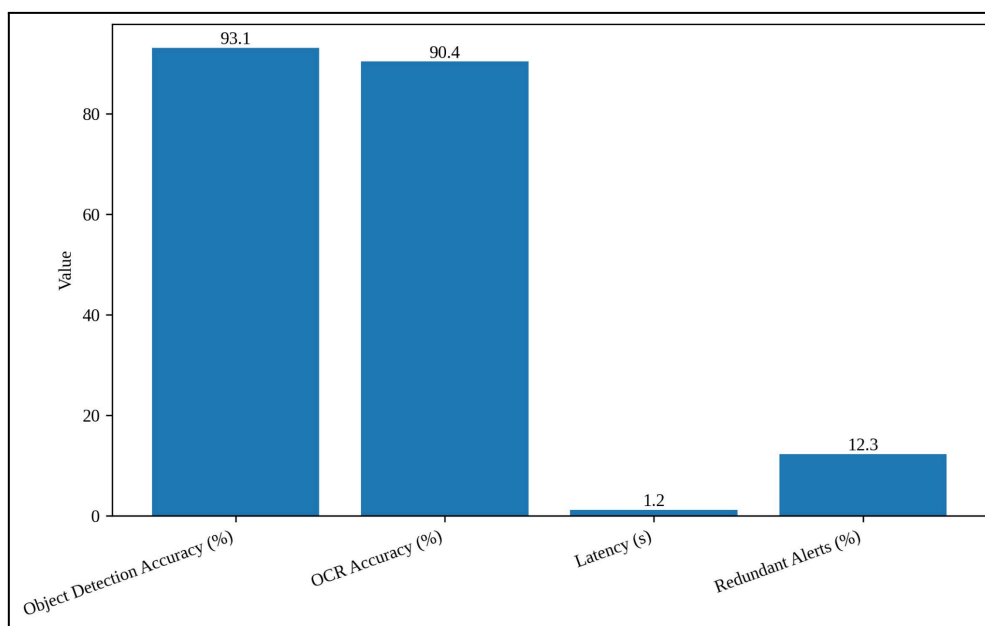


Figure 2. Overall performance evaluation of the proposed AI-based smart glasses system in terms of object detection accuracy, OCR accuracy, system latency, and redundant notification rate.

6.4 Discussion

Finally, experimental data suggest that the proposed smart glasses system is capable of both accuracy, responsiveness, and user friendliness. The high accuracy of object detection (93.1%), strong OCR performance (90.4%) and low latency (1.2 s) relative to the high level of reduction in redundant notifications reflect on the superiority of

the proposed design. These results demonstrate the system's potential for real world application as effective intelligent assistive device for blind people.

7. Conclusion

A study that used an AI-based smart glasses system for real-time assistance to blind or visually impaired users using combined object detection, optical character recognition and speech interaction was demonstrated. Benefiting from a YOLOv5-based perception module, an improved OCR pipeline and a context-aware memory mechanism, our system attained the high retrieval rate for object detection (93.1%), the excellent performance of text recognition (90.4%) as well as the low end-to-end latency (1.2 s) in both indoor and outdoor scenes. The time-based alert suppression helped reduce unnecessary audio alerts, and significantly enhanced the usability and cognitive convenience of the system compared to assistive tools available today. In summary, the experimental results have shown that the developed system provides an effective, scalable and user-friendly approach to improve independent navigation and environmental awareness. In future we aim to include temporal scene understanding, increase the scope of multilingual OCR support, incorporate more environmental sensors as well as optimize for power-efficient runtime on edge hardware to allow sustained real world deployment.

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